📄 AI-BFSI Internship Project Documentation

Solution Requirements – Subscription Waste Detector

1. Introduction

The purpose of this document is to define the **solution requirements** for the Subscription Waste Detector project. It outlines the functional and non-functional requirements, system constraints, and scope of the solution. This ensures clarity for developers, mentors, and stakeholders, while maintaining strict alignment with the implemented features.

2. Project Scope

The Subscription Waste Detector is an **AI-powered web application** designed to help users:

- Upload expense records (CSV, Excel, manual entry).
- Detect recurring subscription payments.
- Categorize subscriptions.
- Visualize subscription spending through dashboards.
- Receive AI-powered cost-saving recommendations via the **Groq API**.

The system operates entirely in-memory and session-based, with no permanent storage or multi-user functionality.

3. Functional Requirements

3.1 Data Input

- Users must be able to upload expense records via:
    - CSV files
    - Excel files
    - Manual entry through the Streamlit interface

3.2 Expense Processing

- The system must:
    - Read and validate uploaded transaction data
    - Display expense records in tabular form
    - Detect recurring subscription payments
    - Categorize subscriptions into meaningful groups

3.3 Dashboard

- The system must generate:
    - Subscription summary

    - Monthly subscription spending overview
    - Interactive charts using Plotly
    - Expense visualization

3.4 AI Features

- The system must:
    - Send processed subscription data to the **Groq API**
    - Display AI-powered cost-saving and optimization recommendations
- **Note:** No custom AI model training is performed.

3.5 Subscription Management

- The system must provide:
    - Renewal reminders
    - Subscription listing

3.6 User Interface

- The system must:
    - Operate as a Streamlit web application
    - Provide an interactive dashboard for user engagement

4. Non-Functional Requirements

- **Performance:**
    - Must process uploaded files and generate dashboards within acceptable response times (<5 seconds for small datasets).
- **Usability:**
    - Simple, intuitive interface using Streamlit components.
    - Clear visualization with Plotly charts.
- **Reliability:**
    - Session-based operation ensures consistent behavior during active use.
- **Security:**
    - No authentication implemented in current version.
    - Data remains in-memory and is not stored permanently.
- **Scalability:**
    - Current implementation supports single-user, session-based operation.
    - Future scalability requires database and authentication integration.

5. System Constraints

- No permanent data storage.
- No multi-user support.
- No integration with external bank APIs.
- No predictive analytics or machine learning training.
- Operates only as a web application via Streamlit.

6. Future Enhancements (Not Implemented Yet)

The following features are planned but not part of the current implementation:

- User authentication and login system
- Database integration (SQLite, Firebase)
- Multi-user support
- Bank statement API integration
- Predictive analytics and spending forecasting
- Multi-currency support
- Cloud synchronization
- Email/SMS notifications

7. Conclusion

The Subscription Waste Detector provides a streamlined solution for identifying and managing subscription expenses. By combining **Python, Pandas, Plotly, and Groq AI**, the system transforms raw expense records into actionable insights.